

Slide 1 - The AWS AI and ML Lifecycle

Hello everyone, and welcome to BLA 1, Part 3 of my AWS Machine Learning Engineer - Associate series. Part 1 introduced AWS cloud foundations, and Part 2 introduced machine learning foundations. This final presentation connects those ideas to the official MLA-C02 exam.

AWS organizes the exam into four domains: data preparation for machine learning and AI; machine learning model and foundation model development; deployment and orchestration of ML and AI workflows; and operating, monitoring, and securing ML and AI solutions. These domains follow the lifecycle of a real system. Data is collected and prepared, a model is selected or developed, the model is deployed through a suitable architecture, and the production solution is monitored, maintained, optimized, and secured.

Slide 2 - MLA-C02 gives nearly equal weight to the full lifecycle

The MLA-C02 exam gives significant weight to every stage of the machine learning lifecycle. Domain 1, Data Preparation for ML and AI, represents twenty-eight percent of the scored content. Domain 2, ML Model and Foundation Model Development, represents twenty-four percent. Domain 3, Deployment and Orchestration of ML and AI Workflows, also represents twenty-four percent. Domain 4, Operating, Monitoring, and Securing ML and AI Solutions, represents the final twenty-four percent.

This distribution communicates an important message. Model development is only one part of machine learning engineering. Data preparation has the largest individual weight, and deployment, operations, monitoring, and security together represent almost half of the scored content. A candidate who studies only algorithms would miss a large portion of the exam.

Slide 3 - Domain 1 turns raw information into trustworthy model input

Domain 1 begins with collecting and storing data. Data can originate from files, relational databases, NoSQL databases, application events, logs, sensors, streams, text, images, audio, and many other sources. AWS services support different access patterns.

After collection, data must be transformed. AWS Glue, Glue DataBrew, Amazon EMR, Apache Spark, Lambda, and SageMaker data tools can support different transformation patterns. Traditional ML preparation can include normalization, standardization, encoding, binning, log transformation, missing-value handling, and outlier detection.

AI preparation also includes embeddings, tokenization, multimodal data handling, and RAG document processing. Documents may be divided into chunks, enriched with metadata, converted to vector representations, and stored in a vector database.

The final responsibility is data quality and bias management. Engineers check completeness, validity, consistency, duplication, class balance, labeling quality, privacy, and representativeness.

Slide 4 - Domain 2 develops predictive models and generative AI

Domain 2 focuses on selecting modeling approaches, training or customizing models, and evaluating performance. Amazon SageMaker AI is a central service for traditional machine learning.

Traditional model development includes controlling overfitting and underfitting, adjusting hyperparameters such as learning rate, batch size, and number of epochs, and applying techniques such as early stopping.

Amazon Bedrock provides managed access to foundation models from AWS and other providers. Engineers select a model according to capability, context size, modality, latency, customization, safety, and price.

Slide 5 - Domain 3 converts a model into a repeatable service

Domain 3 covers deployment infrastructure, resource provisioning, and workflow orchestration. A successful model must be made available in a way that satisfies application requirements. The first decision is the inference pattern. Real-time inference supports interactive requests with low latency.

Engineers also select compute resources, instance families, accelerators, networking, storage, containers, scaling policies, and deployment strategies. A model may have been trained in SageMaker AI or imported from outside AWS. Foundation models may be accessed through Bedrock or deployed through other supported options. RAG systems require vector storage, document indexes, retrieval logic, reranking, and knowledge-base refresh processes. Agents require tools, permissions, state management, and coordination.

Slide 6 - Domain 4 keeps production ML controlled

Domain 4 focuses on what happens after a model or AI workflow enters production. Monitoring is the first responsibility. Engineers observe model quality, data distributions, inference errors, latency, throughput, endpoint health, workflow failures, and infrastructure metrics.

Maintenance responds to what monitoring discovers. A model may need retraining when new data arrives or performance declines.

Cost and performance optimization are also part of operations. Engineers compare instance families, purchasing options, autoscaling settings, capacity models, and inference patterns.

Security applies across the lifecycle. IAM policies should follow least privilege. Data and model artifacts should be encrypted.